

Variables & Equations

ProMo

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1 Variables

2 root

	var	symbol	documentation	type	units	eqs
18	<i>t0</i>	t0	begin time	frame	<i>s</i>	7
5	<i>t</i>	t	time	frame	<i>s</i>	
19	<i>te</i>	te	end time	frame	<i>s</i>	8
4	0.5	onehalf	one half	constant		3
3	1	one	a one	constant		2
1	<i>value</i>	value	a numerical value	constant		
2	0	zero	a zero	constant		1

3 physical

	var	symbol	documentation	type	units	eqs
7	r_{yN}	r_y	y-coordinate	frame	m	
6	r_{xN}	r_x	x-coordinate	frame	m	
8	r_{zN}	r_z	z-coordinate	frame	m	
9	U_N	U	internal energy	state	$kg\,m^2\,s^{-2}$	
12	V_N	V	volume	state	m^3	4
10	S_N	S	entropy	state	$kg\,m^2\,K^{-1}\,s^{-2}$	
11	$n_{N,S}$	n	mass - molar	state	mol	
17	R	R	Gas constant	constant	$kg\,m^2\,mol^{-1}\,K^{-1}\,s^{-2}$	6
16	Avo	Avo	Avogadro number	constant	mol^{-1}	
15	k^B	kb	Boltzmann konstant	constant	$kg\,m^2\,K^{-1}\,s^{-2}$	
20	Axy_N	Axy	cross-sectional area in the x-y plane	secondaryState	m^2	9
22	Ayz_N	Ayz	cross-sectional area in the y-z plane	secondaryState	m^2	11
21	Axz_N	Axz	cross-sectional area in the x-z plane	secondaryState	m^2	10

4 macroscopic

	var	symbol	documentation	type	units	eqs
14	$m_{N,S}$	m	mass	secondaryState	kg	5

5 properties

	var	symbol	documentation	type	units	eqs
13	Mm_S	Mm	molecular mass	constant	$kg\ mol^{-1}$	
23	$density_{N,S}$	density	mass density	secondaryState	$kg\ m^{-3}$	12

6 Equations

7 Generic

no	equation	documentation	layer
1	$0 := \text{Instantiate}(\text{value}, \text{value})$	a zero	root
2	$1 := \text{Instantiate}(\text{value}, \text{value})$	one	root
3	$0.5 := \text{Instantiate}(\text{value}, \text{value})$	one half	root
4	$V_N := r_{xN} \cdot r_{yN} \cdot r_{zN}$	volume	physical
5	$m_{N,S} := \text{properties!}Mm_S \cdot \text{physical!}n_{N,S}$	mass	macroscopic
6	$R := Avo \cdot k^B$	Gas constant	physical
7	$t0 := \text{Instantiate}(t, \text{value})$		root
8	$te := \text{Instantiate}(t, \text{value})$	end time	root
9	$Axy_N := r_{xN} \cdot r_{yN}$	cross-sectional area in the x-y plane	physical
10	$Axz_N := r_{xN} \cdot r_{zN}$	cross-sectional area in the x-z plane	physical
11	$Ayz_N := r_{yN} \cdot r_{zN}$	cross-sectional area in the y-z plane	physical
12	$\text{density}_{N,S} := (\text{physical!}V_N)^{-1} \cdot \text{macroscopic!}m_{N,S}$	mass density	properties